Jian Feng

j36feng@uwaterloo.ca | 647-766-0178 | Website | linkedin/in/fenjian | github/faketut

EMPLOYMENT

Manulife Financial Corporation

Co-op 2026 –

- Built reusable pre-validation & skip-on-failure pattern across ServiceNow catalog task workflows (APIM, AKS, EMU onboarding), eliminating manual error handling and enabling safe async orchestration.
- Migrated SNOW orchestrator from basic auth to GitHub App token authentication via PyGithub, enabling secure mfc-ets team validation with audit-trail compliance across catalog automation.
- Enhanced GOCC orchestrator with Kubernetes command integration for live AKS cert-SAN verification, enabling the Foundry LLM agent to validate gitops PRs against cluster state and catch misconfigs before prod.

Treasury Board of Secretariat

Co-op 2025

- Shipped a .NET MVC module auto-normalizing job titles across a 5K-firm public dataset, unblocking analyst peer-group and governance reporting.
- Migrated 30 Selenium cases (5 suites) to Katalon on Azure DevOps CI/CD; demoed to execs and adopted by other QA teams to ease no-code onboarding.
- Refactored core modules of the enterprise data platform and tuned SQL, cutting 5K-firm-facing browser page load 30% for faster reviews.

Lansi Institute of Artificial Intelligence in Medicine

Co-op (Remote) 2024

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Ran XGBoost + SMR/MR² across a 68-study T2D meta-analysis as research assistant; narrowed the candidate drug-target shortlist used in the published paper.

Sanyang Technology Co., Ltd.

Intern 2024

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Shipped Spring Boot CRUD APIs for cloud reporting/BI modules, unblocking front-end dashboards and downstream ETL consumers.

EDUCATION

University of Waterloo 2023-2028

Bachelor of Computing and Financial Management

- Majors: Computer Science, Finance, Statistics
- Scaled UX Research Hub 32× (10→318 members) as Director; negotiated Google UXR keynote at $0 cost and brokered TikTok chapter cross-promotion for sustained recruitment growth.
- Velocity Co-op Problem Award (top submission among co-op cohort); 3rd Place, SAF Investment Research Challenge; Finalist, SAF Strategy Case Competition
- Certifications: IBM Vector Database, Stanford Machine Learning, UCI Database Design & Operational BI, Rice Parallel-Concurrent-Distributed Programming, UA Reinforcement Learning

PERSONAL PROJECTS

wlp4-compiler

- Multi-pass C++ pipeline: `wlp4scan` → `wlp4parse` (LR) → `wlp4type` → `wlp4gen` emitting ARM64 assembly.
- Codegen lowers control flow, procedures, pointer/array ops with a stable calling convention; transitions traceable via `.wlp4ti` trees.
- −79.6% output size vs the reference `wlp4c` across 65 programs; CI gates every push on the 25-program byte-equal regression suite.

MiniKV

- Go LSM engine — CRC32C WAL, skip-list MemTable, snappy SSTables with sparse index + file-level Bloom so negative lookups skip disk.
- Serializable OCC: snapshot reads, buffered writes, commit-time read-set validation vs a bounded commit log — atomic on no conflict.
- Async leader/follower replication over HTTP NDJSON and gRPC; followers resume from local `LastSeq()` on reconnect, no event replay.

Syncle

- Android spatial-collab client (Kotlin + Compose) renders a 2D office on Canvas with AABB walkability and edge-distance table proximity.
- Realtime over LiveKit: 20 Hz binary position packets, remote-peer lerp, distance-based spatial audio, per-table acoustic isolation.

Anchor

- Splunk drift agent fingerprints a healthy window, then ranks diffs against later windows with LLM-narrated SPL suggestions.
- Feedback loop reweights signals and surfaces recurring unresolved drifts as blind spots.

CabStream ETL

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging → dims/facts → marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

SKILLS

Python, SQL, C#, C++, Java, Git, Go, Kotlin, JavaScript, Linux, Windows, MacOS, .Net, Spring Boot, React, Node.js, Express.js, Flask, PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jira, Confluence, PyTorch, TensorFlow, Scikit-learn, OpenCV, GenAI tools